

# Muhammad Farouk

📍 Cairo, Egypt

📞 +201014332694

✉️ mofasuhu@gmail.com

🌐 linkedin.com/in/muhammadfarouk

🐙 github.com/mofasuhu

🌐 website: mofasuhu.github.io

📷 shutterstock.com/g/MoFarouk

## Python Developer

An innovative and dynamic professional with a BSc in Zoology & Chemistry and a diverse career spanning tutoring, quality assurance, graphics, and software development. Proven expertise in educational technology, data analysis, and content automation with a robust portfolio in academic publishing and e-learning solutions. Passionate about leveraging technology to enhance learning and operational efficiency. Demonstrated success in content management, data analysis, and the automation of tasks within content teams. Adept in Python, Excel, and Google Apps Script to enhance efficiency and foster data-driven decision-making.

## Professional Experience

### Full-Stack E-Commerce Application Developer

Personal Project | Jan 2025 - Present

Designed, developed, and deployed a complete, production-ready e-commerce platform, [HAWARY.SHOP](#), from the ground up using Python, Flask, and PostgreSQL. The application provides a seamless, multilingual shopping experience for customers and a comprehensive dashboard for all business management operations.

#### • Core Application Development:

- Engineered a robust back-end with Python and Flask, featuring secure user authentication (including Google OAuth), session management, and a full-featured RESTful API.
- Built a dynamic, mobile-first front-end using HTML, CSS, and modern JavaScript, ensuring a responsive and accessible user experience across all devices.
- Architected and managed a PostgreSQL database to handle complex relationships between products, clients, orders, and financial transactions, ensuring data integrity and optimized query performance.

#### • E-Commerce & Payment Integration:

- Successfully integrated the Paymob payment gateway API to process secure online credit card transactions, including handling payment callbacks, transaction verification, and processing refunds.
- Developed a complete order fulfillment workflow, allowing admins to manage inventory, update order statuses (Pending, Preparing, Delivered), and track payments for both online and cash-on-delivery orders.
- Implemented a sophisticated shopping cart and promo code system to enhance the customer purchasing journey.

#### • Advanced Feature Implementation:

- Created a comprehensive, secure admin dashboard for end-to-end business management, including multilingual product management (add/update/remove), client account oversight, and order processing.
- Developed a powerful reporting and analytics engine, enabling admins to generate and download detailed sales reports, transaction logs, and customer data for business intelligence.
- Integrated the Google Mail API for automated, event-driven email notifications to customers for order confirmations, status updates, and password resets.

#### • Performance & SEO Optimization:

- Dramatically improved page load times and user experience by implementing server-side pagination, lazy loading for product images, and image optimization (resizing and converting to modern WebP format).
- Enhanced search engine visibility by programmatically generating and serving `sitemap.xml` and `robots.txt` files, ensuring efficient crawling and indexing by Google.
- Implemented server-side caching for translations to deliver a high-performance, fully bilingual (English/Arabic) user interface with right-to-left (RTL) support.

- **Security & Best Practices:**

- Secured the application against common web vulnerabilities by implementing CSRF protection, strict server-side input validation, and environment-based secret key management.
- Ensured all external API communications were conducted over secure (HTTPS) connections with proper SSL/TLS verification.

## **Python Developer (Automation and Data Specialist)**

Nagwa, Cairo, Egypt | Oct 2023 - Present

- Developed and deployed a suite of Python-based tools to automate complex data processing and content management tasks, significantly enhancing operational efficiency and reducing manual effort for content teams.
- Engineered a robust data migration pipeline to process over 600,000 JSON files. This involved creating sophisticated Python scripts for data categorization, cleaning, and transformation, including:
  - Parsing XML files to replace SVG elements with corresponding LaTeX code in JSON.
  - Converting LaTeX commands into MathLive syntax to support a modern, interactive content standard.
  - Implementing a YAML-based regex management system for precise, large-scale text and code replacements.
  - Developing specialized functions to handle edge cases and ensure high-fidelity data conversion.
- Led data-driven initiatives by performing large-scale data extraction, cleaning, and analysis using Python, enabling informed strategic decision-making.
- Authored and optimized scalable Python scripts for efficiency, ensuring seamless processing of large datasets and automating critical file management tasks.
- Contributed to open-source by publishing selected Python automation tools. A portfolio of projects is available on my GitHub: [github.com/mofasuhu](https://github.com/mofasuhu).
- Ensured data integrity throughout the migration and transformation processes by implementing rigorous testing, debugging, and validation protocols.

## **Graphics Quality Assurance Specialist, Newcomers Trainer, and Python Developer**

Nagwa, Cairo, Egypt | Jan 2017 - Sep 2023

- Devised innovative drawing techniques utilizing LaTeX, Chemdoodle, Adobe Illustrator, and Adobe Photoshop.
- Orchestrated comprehensive training programs for newcomers, monitoring progress and providing ongoing support.

## **Quality Assurance Specialist**

Egypt Man Co. for Food Industries, Cairo, Egypt | Feb 2015 - Jun 2016

- Oversaw quality control processes and material standards, ensuring adherence to company and industry guidelines.
- Produced technical files to facilitate clear communication and project documentation.

## **Finalization Quality Assurance Specialist**

Hindawi Publishing, Cairo, Egypt | Sep 2012 - Feb 2015

- Conducted rigorous quality checks to ensure published materials aligned with in-house journal styles and standards.
- Generated HTML and ePub versions of articles, enhancing accessibility and reader engagement.

## **Medical Customer Service Representative**

Wasla Outsourcing, Cairo, Egypt | Mar 2011 - Jun 2012

- Managed patient inquiries with empathy and efficiency, maintaining high standards of customer service and care.

## **Science Teacher**

Talaea Al-Azhar Al-Sharief Institute, Cairo, Egypt | Sep 2009 - Aug 2010

- Delivered engaging science lessons to prep-school students, fostering a love of learning and scientific inquiry.

## Education

Bachelor of Science (B.Sc.) in Zoology and Chemistry

Ain Shams University, Cairo, Egypt | 2009

## Certifications and Courses

- Data Analysis Challenger Track - Udacity
- Data Analysis Professional Track - Udacity
- Learning Django - LinkedIn Learning
- Learning Git and GitHub - LinkedIn Learning
- Tableau Essential Training - LinkedIn Learning
- Learning REST APIs - LinkedIn Learning
- Django Essential Training - LinkedIn Learning
- Django: Forms - LinkedIn Learning
- Building RESTful APIs with Flask - LinkedIn Learning
- Introduction to AWS for Non-Engineers - LinkedIn Learning
- Python Essential Training - LinkedIn Learning
- Building a Personal Portfolio with Django - LinkedIn Learning
- Deploying Django Apps - LinkedIn Learning
- Become a Django Developer - LinkedIn Learning
- English for Career Development Self-Paced MOOC 2022 - Canvas Credentials (Badgr)
- Water analysis principles and applications - Ain Shams University
- Principles and applications of advanced chromatographic techniques - Ain Shams University
- Basic Business Skills Acquisition Program "BBSA" - AMIDEAST, CAME, Global Knowledge
- TOEIC - Score: 585 - Amideast
- TOEFL - Score: 537 - Amideast

## Skills

### Technical

Python

Coding

Google AppScript

Automation

Data Processing and data analysis

Flask

Django

AutoCAD

Adobe Illustrator

Chemdoodle

Tableau

Git and GitHub

### Soft Skills

Creative problem-solving

Analytical thinking

Teamwork

Academic publishing

Educational technology

## References

To view the certificates, please click 